NARAYANA
IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | GTM-21 | Date: 20.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

# KEYSHEET

## PHYSICS

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| B | C | A | B | A | B | D | A | A | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | D | C | D | B | B | B | D | A | A |
| 21 | 22 | 23 | 24 | 25 | | | | | |
| 6 | 5 | 5 | 1 | 2 | | | | | |

## CHEMISTRY

| 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 |
|---|---|---|---|---|---|---|---|---|---|
| B | D | C | C | A | D | D | C | A | D |
| 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 |
| D | B | C | C | A | C | B | C | D | C |
| 46 | 47 | 48 | 49 | 50 | | | | | |
| 12 | 50 | 7 | 1005 | 103 | | | | | |

## MATHEMATICS

| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
|---|---|---|---|---|---|---|---|---|---|
| C | B | C | C | C | B | D | A | D | D |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| A | B | D | A | B | B | D | C | C | A |
| 71 | 72 | 73 | 74 | 75 | | | | | |
| 83 | 50 | 1001 | 18 | 2010 | | | | | |



For More Material Join: @JEEAdvanced_2025

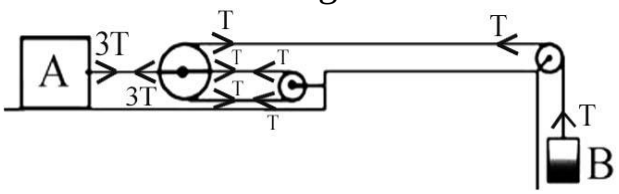
## PHYSICS

1. Pressure gradient = $\frac{dp}{dx} = \frac{ML^{-1}T^{-2}}{L} = ML^{-2}T^{-2}$

   Work function $(\phi)$ = Energy $= ML^2T^{-2}$

   Moment of inertia $= Ml^2 = ML^2$

   Electric field $= \frac{Force}{Charge} = \frac{MLT^{-2}}{AT} = MLT^{-3}A^{-1}$

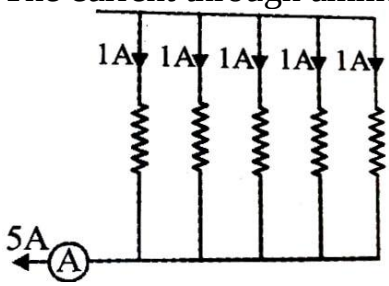
2. 


   $3T.V_A - T.V_B = 0$

   $3V_A = V_B \rightarrow V_B = 3\times 0.6$

   $= 1.8\, m/s$

3. The current through ammeter $i = 5A$




   $R_e = \frac{9}{9} = 1\Omega$

   $i = \frac{V}{R_e} = \frac{9}{1} = 9A$

   Current through each resistor = 1A

   $\therefore$ ammeter reading = 5A

4. As, $e = \frac{L^2 dg}{2Y} = \frac{\left(8\times 10^{-2}\right)^2 \times 1.5\times 10}{2\times 5\times 10^8} = 9.6\times 10^{-11} m$

5. Here, $\lambda_0 = 3800\,\text{Å}$

   Work function $\phi$ (or)

   $W = h\nu_0 = \frac{hc}{\lambda_0} = \frac{6.63\times 10^{-34}\times 3\times 10^8}{3800\times 10^{-10}}$

   $= 5.23\times 10^{-19} J$

6. $P_i = P_t$

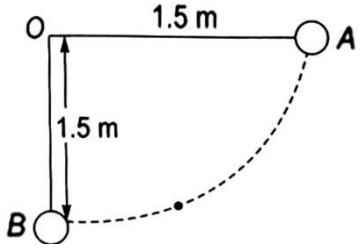
   $\therefore mv = (100m)V_f$

   or $V_f = \frac{V}{100}$

7. Length of the pendulum = 1.5m



For More Material Join: @JEEAdvanced_2025




Potential energy of the bob at position A = mgh

As bob moves from position A towards position B its potential energy converted into kinetic energy. 5% of its potential energy is dissipated against air resistance

$\therefore$ KE at position B = 95% of its PE at position A

$\frac{1}{2}mv^2 = \frac{95}{100} \times mgh$ (or)

$v = \sqrt{\frac{2 \times 95 \times gh}{100}}$

$= \sqrt{\frac{19}{10} \times 9.8 \times 1.5} = 5.28\, m/s$

8. Speed of light in vacuum, $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ .....(i)

And in another medium $v = \frac{1}{\sqrt{\mu \varepsilon}}$ .....(ii)

Dividing Eq. (i) by (ii), we get $\frac{C}{V} = \sqrt{\frac{\mu}{\mu_0} . \frac{\varepsilon}{\varepsilon_0}}$

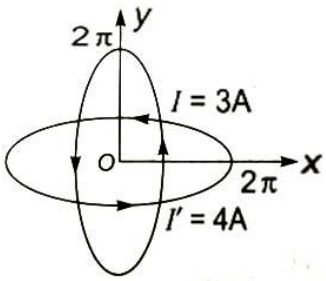
$V = C . \sqrt{\frac{\mu_0 \varepsilon_0}{\mu \varepsilon}}$

$V = \frac{C}{\sqrt{\mu k}}$

Where $\mu$ = ratio of two permeabilities and K = dielectric constant (ratio of two permittivities)

9. Field at the centre of the loop is given by $B = \frac{\mu_0 I}{2R}$

Where, R is radius given, $R = 2\pi\ cm = 2\pi \times 10^{-2} m$




$B_x = \frac{\mu_0}{2} \frac{l_1}{2\pi \times 10^{-2}}$

$l_1 = 3A$



For More Material Join: @JEEAdvanced_2025

$$\therefore B_x = \frac{\mu_0}{2}.\frac{3\times10^2}{2\pi} = 3\times10^{-5}T$$

$$B_y = \frac{\mu_0}{2}.\frac{l_2}{2\pi\times10^{-2}}, l_2 = 4A$$

$$B_y = 4\times10^{-5}$$

$$\therefore B_{net} = \sqrt{B_x^2 + B_y^2} = \sqrt{\left(3^2+4^2\right)\times10^{-10}}$$

$$B_{net} = 5\times10^{-5}T$$

10. $a\sin\theta = n\lambda$

$$\sin\theta = \frac{1.\lambda}{a}$$

$\therefore$ width of the slit is

$$a = \frac{\lambda}{\sin 30}$$

$$a = 2\times6000\times10^{-10}$$

$$= 1.2\times10^{-6}M$$

11. Here, $N_P = 400, N_S = 300$

$E_S = 300, E_P = ?$

Using $\frac{E_S}{E_P} = \frac{N_S}{N_P}$, we get

$$E_P = \frac{N_P}{N_S}\times E_S = \frac{400\times300}{300} = 400V$$

12. The gravitational potential at the point P due to the particle at the centre is

$$V_1 = -\frac{GM}{a/2} = -\frac{2GM}{a}$$

The potential at P due to the shell is $V_2 = -\frac{GM}{a}$

The net potential at P is $V_1 + V_2 = -\frac{3GM}{a}$

13. Given, resistance of silver wire at 27.5°C $= R_{27.5} = 2.1\Omega$

Resistance of silver wire at 100°C = $R_{100} = 2.7\Omega$

Let the temperature coefficient of silver be $\alpha$

$$\alpha = \frac{R_{t_2} - R_{t_1}}{R_1\left(t_2 - t_1\right)}$$

$$\alpha = \frac{R_{100} - R_{27.5}}{R_{27.5}\left(100 - 27.5\right)} = \frac{2.7-2.1}{2.1\times72.5}$$

$$\alpha = 0.0039/°C$$

Thus, the temperature coefficient of resistivity of silver is $0.0039/°C$

14. $V_Z = 5V$

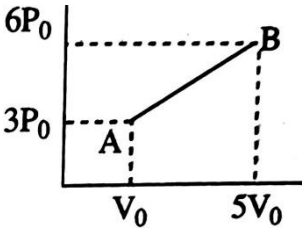
$$I_L = \frac{V_Z}{R_L} = \frac{5}{10^3} = 5mA$$



For More Material Join: @JEEAdvanced_2025

15. $dT = T_B - T_A = \frac{P_B V_B}{R} - \frac{P_A V_A}{R}$

$= \frac{6P_0 5V_0}{R} - \frac{3P_0 V_0}{R} = \frac{27P_0 V_0}{R}$




$dW = \frac{1}{2}(3P_0 + 6P_0)(4V_0) = 18P_0V_0$

$\frac{R}{1-x} = \frac{dW}{(T_2 - T_1)} = \frac{18P_0V_0(R)}{27P_0V_0} = \frac{2}{3}R$ [In polytropic process $\because dW = \frac{R}{1-x}(T_2 - T_1)$]

According to polytropic process

Sp. Heat, $C = C_V + \frac{R}{1-x}$

Sp. Heat $= \frac{3}{2}R + \frac{2}{3}R = \frac{13R}{6}$

16. $E_{axis} = \frac{2kp}{r^3}$ and $E_{\text{bisector}} = \frac{kp}{(2r)^3}$

$E_2 = \frac{kp}{8l^3} = \frac{2kp}{16l^3} = \frac{E_1}{16} \Rightarrow \overrightarrow{E_2} = -\frac{\overrightarrow{E_1}}{16}$ [$\because \overrightarrow{E_2}$ is antiparallel to $\overrightarrow{E_1}$]

17. Only $\hat{k}$ component of magnetic field will produce the motional emf.

$e = BVl = (5)(5)(t) = 25V$

18. $\vec{L} = m(\vec{r} \times \vec{v}) = 50\begin{vmatrix} \hat{i} & -\hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$

$= 50\left[(15-16)\hat{i} - \hat{j}(10-12) + \hat{k}(8-9)\right]$

$= -50\hat{i} + 100\hat{j} - 50\hat{k}$

19. $v_{rms} = \sqrt{\frac{3RT}{M}} \Rightarrow v_{rms} \propto \sqrt{T}$

$\Rightarrow \frac{V}{(V/2)} = \sqrt{\frac{(273+327)}{T_2}} \Rightarrow 4 = \frac{600}{T_2}$

$T_2 = 150k \Rightarrow t = 150 - 273 = -123°C$

20. At frequency $\omega, X_C = 1/\omega C$

At frequency $\omega/3, X'_C = \frac{3}{\omega C} = 3X_C$



For More Material Join: @JEEAdvanced_2025

$$I=\frac{V}{\sqrt{R^2+X_C^{\ 2}}};\frac{I}{2}=\frac{V}{\sqrt{R^2+9X_C^{\ 2}}};$$

$$\Rightarrow\frac{I}{I/2}=\frac{\sqrt{R^2+9X_{c^2}}}{\sqrt{R^2+X_{c^2}}}\Rightarrow 4R^2+4X_{c^2}=R^2+9X_{c^2}$$

$$3R^2=5X_{c^2}$$

$$\frac{X_C}{R}=\sqrt{\frac{3}{5}}$$

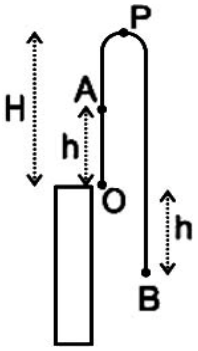
21. $$m=\frac{\mathrm{f}}{\mathrm{f}+u}$$

$$m_{25}=\frac{20}{20-25}=-4$$

$$m_{50}=\frac{20}{20-50}=\frac{-2}{3}$$

$$\frac{m_{25}}{m_{50}}=\frac{-4}{(-2/3)}=6$$

22.




Given that $\mathrm{v}_B=2\mathrm{v}_A$

$$\mathrm{v}_B^2=4\mathrm{v}_A^2$$

$$\mathrm{v}_0^2+2gh=4\left(\mathrm{v}_0^2-2gh\right)$$

$$10gh=3\mathrm{v}_0^2$$

$$\mathrm{v}_0^2=\frac{10gh}{3}$$

$$\mathrm{v}_p^2-\mathrm{v}_0^2=2(-g)H$$

At highest point velocity is zero

$$\mathrm{v}_p=0$$

So, $H=\frac{\mathrm{v}_0^{\ 2}}{2g}$

$$=\frac{10gh/3}{2g}$$

$$H=\frac{5h}{3}$$



For More Material Join: @JEEAdvanced_2025

23. Common potential $= \dfrac{C_1V_1 \sim C_2V_2}{C_1 + C_2}$

$V_C = \dfrac{C.V \sim 2C.2V}{C + 2C} = V$

Final energy $= \dfrac{1}{2}(C_1 + C_2)V_C{}^2$

$= \dfrac{1}{2}(C + 2C)V^2$

$= \dfrac{3}{2}CV^2$

$x = 3,\ y = 2$

$x + y = 3 + 2 = 5$

24. The frequencies are in the ratio of 5 : 7 : 9

Hence, it is a closed pipe

$425Hz = 5\left(\dfrac{\mathrm{v}}{4l}\right)$ (or) $l = \dfrac{5\mathrm{v}}{4 \times 425} = \dfrac{5 \times 340}{4 \times 425} = 1m$

25. $e = \dfrac{1}{2}BR^2\omega$ where $\omega = 1200 \times \dfrac{2\pi}{60} = 40\pi$

$e = \dfrac{1}{2} \times 10^{-2} \times (0.1)^2 \times (40\pi)$

$= \dfrac{4\pi}{2} \times 10^{-3}$

$= 2\pi MV$



For More Material Join: @JEEAdvanced_2025

26. $4^{th}$ option is an aldol product but option (2) is from benzaldehyde and acetophenone, option (1) is Cannizzaro product

27. $\frac{t_{50\%}}{t_{20\%}} = \frac{\log 2}{\log \frac{100}{80}} = \frac{0.3010}{1-3(0.301)} = \frac{0.301}{1-0.903} = \frac{0.301}{0.097} \simeq 3.1$

$\therefore t_{50\%} = 3.1 \times 20 = 62 \text{ min}$ (or)

$100 \xrightarrow[20\%]{-20} 80 \xrightarrow[20\%]{-16} 64 \xrightarrow[20\%]{12.8} 51.2 \longrightarrow 50$

$\therefore$ Total time $20+20+20+t => 60$

28. Formation of 'C' is due to Hoffman hypobromite reaction which is not a possible product ($2^o$-amine)

29. Option (2) and (4) are aromatic while (3) is anti-aromatic highly unstable

30. $Nd^{+3} \rightarrow [Xe]6s^0 5d^0 4f^3$

$Eu^{+3} \rightarrow [Xe]6s^0 5d^0 4f^6$

$Yb^{+2} \rightarrow [Xe]6s^0 5d^0 4f^{14}$ (no unpaired)

31. Violet colored complex ion is $\left[Fe(CN)_5 NOS\right]^{-4}$, where coordination number is six, whose geometry is octahedral

32. Diazonium salt is stabilized by benzene ring (resonance)

33. Conversion of a carbonyl compound into corresponding alkane in acidic medium is Clemenson reduction and in basic medium is Wolff-Kishner reduction

34. $A^{+2} + 2e^- \rightarrow A \qquad E_1^0 = -0.56V$

$A^{+2} \rightarrow e^- + A^{+3} \qquad E_2^0 = -1.48V \times 2$

$3A^{+2} \rightarrow A + 2A^{+3} \qquad E_{cell}^0 = E_{cathode} - E_{anode} (only\ SRP)$

$= -0.56 - (1.48)$

$= -2.04V$

35. 'A' is a ketone $C_6H_5 - CH_2 - \overset{O}{\overset{\|}{C}} - CH_3$. It can give iodoform test. 'B' is formed by Etard reaction

36. $sp^3$ carbon adjacent to double bonded carbon is allylic carbon

37. $I_3^{-1}$ central atom has 3 lone pairs, 2 bond pairs

$NO_3^-$ central atom nitrogen has no lone pairs, 3 bond pairs

$D_2O$ similar to water molecule (2 lone pairs + 2 bond pairs)

$PCl_5$ (No lone pairs only 5 bond pairs)

38. Longest carbon chain should include the carbon atoms having functional groups, ignoring the longest chain rule

39. $HC \equiv C^{\Theta} + C_2H_5 - Cl \longrightarrow HC \equiv C - CH_2 - CH_3$ (n-butyne)

$C^{\Theta} \equiv C^{\Theta} + 2CH_3Cl \longrightarrow CH_3 - C \equiv C - CH_3$ (2-butyne)



For More Material Join: @JEEAdvanced_2025

40. Given element is Livermorium (116) with atomic number = 116

41. Ratio for the gap = 3 : 5 : 7 but not 1 : 3 : 5

43. Iron in the above complex is $+1$ but not $+3$

44. Both sugars are reducing sugars due to the presence of hemi-acetal group

45. Sulphite + dil. Acid $\longrightarrow SO_2$, ($SO_2$ turns dichromate solution into green)

46. No. of millimoles (I) $= V_1 \times M_1 = 50$

No. of millimoles (II) $= V_2 \times M_2 = 50$

$\therefore$ added $NaOH = y$ millimoles

$$\therefore M = \frac{n}{V} = \frac{50+50+y}{1000} = 0.4 \Rightarrow y = 300 \text{ millimoles}$$

$$\therefore x = 300 \times 10^{-3} \times 40\, g$$

$$= 12\, g$$

47. Ferric ferro cyanide $Fe_4\left[Fe(CN)_6\right]_3$

$x = 7 \quad y = 43$

48. $Cr_2O_7^{-2} + 14H^+ + 3C_2O_4^{-2} \longrightarrow 2Cr^{3+} + 6CO_2 + 7H_2O$

$x = 1 \qquad y = 3 \qquad z = 6$

$\therefore x + z = 1 + 6 = 7$

49. $\Delta T_b = K_b \times m = 0.5 \times 0.2 = 0.1K \Rightarrow T_b^1 = 373 + 0.1$

$\Delta T_f = K_f \times m = 2 \times 0.2 = 0.4K \Rightarrow T_f^1 = 273 - 0.40 = 272.6$

$\therefore T_b^1 - T_f^1 = 373.1 - 272.6 = 100.5K$

50. $K_{sp} = 4S^3 = 4 \times 10^{-12} \Rightarrow S = 10^{-4} \Rightarrow \left(OH^-\right) = 2S = 2 \times 10^{-4}$

$\Rightarrow P^{OH} = 4 - \log 2 \Rightarrow P^H = 10 + \log 2 = 10.3$



For More Material Join: @JEEAdvanced_2025

51. Line $L_1$ also passes through the point $(-1,0)$

$\therefore$ slope of $L_2$ is $\tan 60° = \sqrt{3}$

52. Use $T_n = S_n - S_{n-1}$

First A.P is 3, 7, 11, …… Second A.P is 2, 5, 8, …….

C.D of common terms series is LCM of C.D's of two A.P's

53. $9iz^2(3z-2i)-4(3z-2i)=0$

$\therefore (3z-2i)(9iz^2-4)=0$

54. $(-\bar{a}+2\bar{b})\times\bar{c}=\bar{0}$

$\therefore \bar{c}=t(5\bar{i}+\bar{j}-3\bar{k})$

$(3\bar{a}+\bar{b}).\bar{c}=42 \Rightarrow t=2$

55. $h^2+r^2=27$ and $V=\frac{1}{3}\pi r^2 h=\frac{\pi h}{3}(27-h^2)$

$\frac{dv}{dh}=0 \Rightarrow h=3$

56. Last term $=\frac{2n}{n^2}\sin\left[\frac{(2n)^2+n^2}{n^2}\right]$

$\therefore \underset{n\to\infty}{Lt}\frac{1}{n}\sum_{r=1}^{2n}\frac{r}{h}\sin\left(1+\frac{r^2}{n^2}\right)=\int_0^2 x\sin(1+x^2)dx$

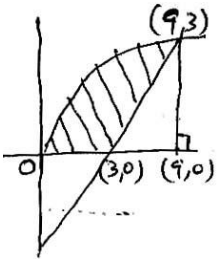
57. Domain is $[2,\infty)$ and $\sqrt{x-2\sqrt{2(x-2)}}=\sqrt{2}-\sqrt{x-2}$ for $2\le x<4$

$=\sqrt{x-2}-\sqrt{2}$ for $x\ge 4$

$\therefore f(x)=2\sqrt{2}$ for $2\le x<4$

$=2\sqrt{x-2}$ for $x\ge 4$

58. Shaded area $=\int_0^9 \sqrt{x}\,dx-\frac{1}{2}(6)(3)$




59. $\frac{dy}{dx}=\frac{x+\sin x}{1+\cos x}=\frac{x}{2}\sec^2\frac{x}{2}+\tan\frac{x}{2}$

Integrating, $y(x)=x\tan\frac{x}{2}+c$

60. Let foot of perpendicular $Q=(2k+1,-3k-1,8k-10)$ using PQ perpendicular to line, k = 1

If R is image, mid-point of PR is Q



For More Material Join: @JEEAdvanced_2025

61. $0,1,1,10,11 \rightarrow {}^5C_2.{}^3C_1.{}^2C_1.\left(\frac{1}{4}\right)^5$

$0,0,1,11,11 \rightarrow {}^5C_2.{}^3C_2.\left(\frac{1}{4}\right)^5$

$1,1,1,10,10 \rightarrow {}^5C_2.\left(\frac{1}{4}\right)^5$

adding, probability $=\left(\frac{1}{4}\right)^5(100)$

62. $-1 \le \frac{x}{2}-3 \le 1 \Rightarrow 4 \le x \le 8$

$49-x^2 \ge 0 \Rightarrow -7 \le x \le 7$

$x^2-11x+30>0 \Rightarrow x<5$ (or) $x>6$

Domain is $[4,5) \cup (6,7]$

63. After cancelling $x$, use $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

64. $r_1-r_2<AB<r_1+r_2$

$\therefore n_1=2, n_2=0$

65. $\frac{2b^2}{a}=ae \Rightarrow 2e^2+e-2=0$

66. Median (M) = Value of $x$ corresponding to c.f just greater than $\frac{N}{2}=25$ is 13

$\therefore M.D=\frac{1}{N}\sum f(x-M)=\frac{1}{50}(136)$

67. $c_1 \rightarrow c_1-(2\sin x)c_3$; $c_2 \rightarrow c_2+(2\cos x)c_3$ and expand. Then $\Delta=2$

68. $\Delta_1=\left|\bar{a}\times\bar{b}\right|=\sqrt{41}$, $\Delta_2=\frac{1}{2}\left|\bar{p}\times\bar{q}\right|=5\sqrt{3}$

69. $2(\sec^2 x-1)-5\sec x=1$ gives $\sec x=3$ has 7 solutions in

$\left[3(2\pi)+\frac{\pi}{2}, 3(2\pi)+\frac{3\pi}{2}\right]$

70. Using L-Hospital rule and Leibnitz formula, limit value = 1

$\therefore \alpha=\beta+1, \beta=2\alpha-1$

71. Using formula, p = 9

Foot of perpendicular on second line is $(1,3,-1)$

$\therefore q=\sqrt{2}$

72. $2x+3=t$ gives $\frac{1}{2}\int_{-5}^{5}\|[t]\|\,dt=\frac{1}{2}\left[5+(4+3+2+1)2\right]=\frac{25}{2}=\alpha$

73. $T_{r+1}={}^{1000}C_r.2^{\frac{1000-r}{3}}.5^{\frac{r}{6}}$



For More Material Join: @JEEAdvanced_2025

For the term to be rational, r is multiple of 6 and $(1000-r)$ is multiple of 3. But $\frac{1000-r}{3}=\frac{1000}{3}-\frac{6k}{3}=\frac{1000}{3}-2k$ which cannot be multiple of 3

74. $\frac{dy}{dx}-\frac{x-2}{x(x-1)}y=\frac{x^3(2x-1)}{x(x-1)}$

$\frac{x-2}{x(x-1)}=\frac{2}{x}-\frac{1}{x-1}$ $\qquad I.F=e^{\int\left(\frac{1}{x-1}-\frac{2}{x}\right)dx}=\frac{x-1}{x^2}$

Solution is $y.\frac{x-1}{x^2}=\int(2x-1)dx=x^2-x+c$

$y(2)=0\Rightarrow C=-2$ $\qquad \therefore y(3)=18$

75. $10^{th}$ bracket contains 20 terms with c.d 1

$1^{st}$ terms of brackets are 1, 3, 7, 13 ……. Whose $10^{th}$ term is

$1+(2+4+6+.....9\text{ terms})=$

$1+2\frac{(9)(10)}{2}=91$

$\therefore S_{20}=\frac{20}{2}\left[2(91)+(20-1)1\right]$



For More Material Join: @JEEAdvanced_2025